



SEMESTER 2 EXAMINATIONS 2022/2023

MODULE: CA216 - Operating Systems

PROGRAMME(S):

COMSCI	BSc in Computer Science
ECSAO	Study Abroad (Engineering & Computing)
CASE	BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY: 2,0

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1

[TOTAL MARKS: 25]

Q 1(a)

[8 Marks]

Provide a labelled diagram of the Linux process life-cycle with explanations i.e., the states a process can be in and how it can transition between these states.

Q 1(b)

[9 Marks]

Explain using example C code the memory layout of a C program. In your C code, provide an example of what is stored in each of the following memory sections: *text*, *initialised data*, *uninitialised data*, *heap* and *stack*.

Q 1(c)

[4 Marks]

Describe an advantage of using threads compared to forking separate processes.

Q 1(d)

[4 Marks]

Briefly explain the purpose of the following two commands on Linux:

- a) `strace`
- b) `kill`

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[7 Marks]

Using Amdahl's law, calculate the speedup of a program with a 75% serial proportion for 10 processors. Assuming there was an unlimited number of processors available, what's the maximum speedup that could be achieved?

Q 2(b)

[7 Marks]

Explain the purpose of the `test_and_set` OR `compare_and_swap` processor instructions using an example.

Q 2(c)

[7 Marks]

Outline a solution to the producer consumer problem using C code using semaphores. You may assume the existence of semaphores (with notation *sem*) and related operations (i.e., `wait()` and `signal()`).

Q 2(d)**[4 Marks]**

An operating system is using a shortest-remaining-time-first scheduler i.e., the pre-emptive version of the SJF (Shortest Job First).

The following lists of tasks arrive for scheduling:

Process	Arrival Time	Burst Time
A	0	2
B	1	2
C	4	5
D	7	2
E	9	4

What will be the **waiting time** of Process B?

[End of Question 2]

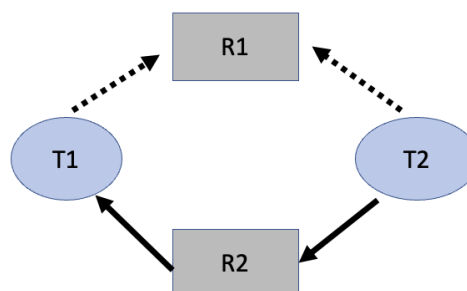
QUESTION 3**[TOTAL MARKS: 25]****Q 3(a)****[9 Marks]**

Explain the concept of deadlock using an example. In your answer describe the necessary requirements for deadlock to occur.

Q 3(b)**[5 Marks]**

Consider the resource-allocation graph below. Both T1 and T2 request resource R1 at the same time. As there is only a single instance of R1 available, only one of these requests can be converted into an assignment. **Note:** A broken-arrow line indicates a claim edge. R1 and R2 are resources for which there is only a single instance available for each. T1/T2 are tasks/processes that need to use these resources to complete their execution.

Should R1 be assigned to T1 or T2 first? **Explain your answer.**



Q 3(c)**[7 Marks]**

A programmer is attempting to implement a solution to first readers-writers problem but has encountered difficulty when using the implementation below. Two semaphores **rw_mutex** and **mutex** which have both been initialised to **1**, and an integer **read_count** initialised to 0, are available to both writer and reader processes/threads.

Outline what changes you would make to the implementation below for it to operate correctly. In your answer, refer to specific line numbers.

Writer process:	Reader Process:
<pre>01: while (true) { 02: wait(rw_mutex); 03: ... 04: /* writing is performed */ 05: ... 06: signal(mutex); 07: }</pre>	<pre>08: while (true){ 09: signal(rw_mutex); 10: read_count--; 11: if (read_count == 0) { 12: wait(rw_mutex); 13: } 14: signal(mutex); 15: ... 16: /* reading is performed */ 17: ... 18: wait(mutex); 19: read_count++; 20: 21: if (read_count < 0) { 22: signal(rw_mutex); 23: } 24: signal(rw_mutex); 25: }</pre>

Q 3(d)**[4 Marks]**

Is the following statement true or false? Explain why or why not?

Race conditions are only possible on multi-processor/multi-core computers.

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[4 Marks]**

I run the following sequence of commands on Linux:

```
echo "123" > myfile1.txt  
ln myfile1.txt myfile2.txt  
rm myfile1.txt  
echo "XYZ" > myfile1.txt  
ln myfile1.txt myfile3.txt  
echo "ABC" > myfile3.txt
```

What (if anything) is contained in myfile2.txt?

Q 4(b)**[4 Marks]**

In the context of filesystems, briefly explain the difference between hard and soft links.

Q 4(c)**[7 Marks]**

Explain the purpose of the Translation Lookaside Buffer (TLB) in Paging. What would be the impact on performance if this was not used?

Q 4(d)**[7 Marks]**

Using a 32-bit or 64-bit virtual address as an example, describe the implications of using a small versus a large frame/page size. What might be the negative impacts of having a very small or very large page/frame size? Explain your answer.

Q 4(d)**[3 Marks]**

Explain what the following Linux command does:
chmod 655 myfile

[End of Question 4]**[END OF EXAM]**